

Fill in the blanks



Based on the provided table and the context of scheduling ring all-reduce learning jobs, the missing value for the "Size of each message" column should be consistent with the pattern observed. The "All-to-All" method has a size of p , and the "Parameter Server" and "Ring All-Reduce" methods do not specify a size, suggesting it might be the same or a related variable. Given the typical conventions in distributed computing where message size is often a parameter (e.g., p for data size), and without additional context to suggest otherwise, the most reasonable assumption is that the "Size of each message" for "Parameter Server" and "Ring All-Reduce" is also p .

Here is the completed table:

	Number of Messages Sent	Size of each message
All-to-All		p
Parameter Server	$2n$	p
Ring All-Reduce	$n(2n - 1)$	p